Taeseong Kim

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Overview

I am an Integrated Ph.D. student at the Department of Mathematical Sciences, Seoul National University (SNU), Republic of Korea. My advisor is Prof. Jung Hee Cheon. I work on cryptography, currently focusing on Homomorphic Encryption (HE) and its applications, including privacy-preserving machine learning.

Publications

Authors are listed in alphabetical order by last name.

Conference Publications

Youngjin Bae, Jung Hee Cheon, Minsik Kang, Taeseong Kim‡ — *WAHC*

[C03] High-Throughput AES Transciphering using CKKS: Less than 1ms — *2025*

‡ denotes corresponding author

Dongwhan Rho†**, Taeseong Kim**†**, Minje Park, Jung Woo Kim, Hyunsik Chae, Jung Hee Cheon, Ernest K. Ryu** — *ICLR*

[C02] Encryption-Friendly LLM Architecture — *2025*

† denotes co-first author

Wonhee Cho, Guillaume Hanrot, Taeseong Kim, Minje Park, Damien Stehlé — *ACM CCS*

[C01] Fast and Accurate Homomorphic Softmax Evaluation — *2024*

Journal Publications

Jung Hee Cheon, Minsik Kang, Taeseong Kim, Junyoung Jung, Yongdong Yeo — *IEEE TDSC*

[J01] Batch Inference on Deep CNNs With Fully Homomorphic Encryption Using Channel-By-Channel Convolutions — *Aug. 2024*

Manuscripts

Jung Hee Cheon, Seungwan Hong, Minsik Kang, Jonghyun Kim, Taeseong Kim, Changmin Lee, Junho Lee — *IACR ePrint 2026/853*

[M01] MRFHE: Mixed-Radix Fully Homomorphic Encryption with Better Batch Bootstrapping — *2026*

Taeseong Kim, Jonghoo Lee, Taeyeong Noh, Jung Hee Cheon, Guillaume Hanrot — *IACR ePrint 2026/1209*

[M02] Latency-Aware, High-Throughput Homomorphic AES Evaluation with CKKS — *2026*

Jung Hee Cheon, Kevin Choi, Hyeoncheol Joo, Taeseong Kim — *IACR ePrint 2026/2360*

[M03] One-Round Threshold XMSS and SPHINCS+ from Fully Homomorphic Encryption — *2026*

Talks

2026

PACOH Workshop: Homomorphic Encryption 2026 — *Sokcho, South Korea*

Latency-Aware, High-Throughput Homomorphic AES Evaluation with CKKS — *June. 2026*

2025

Korea Mathematics Society Spring Meeting — *Daejeon, South Korea*

High-Throughput AES Transciphering using CKKS: Less than 1ms — *Apr. 2025*

2024

Korea Mathematics Society Fall Meeting — *Suwon, South Korea*

Encryption-Friendly LLM Architecture — *Oct. 2024*

Korea Mathematics Society Spring Meeting — *Daejeon, South Korea*

Fast and Accurate Homomorphic Softmax Evaluation — *May 2024*

2023

Korea Mathematics Society Fall Meeting *Seoul, South Korea*

Batch Inference on Deep Convolutional Neural Networks With Fully Homomorphic Encryption Using Channel-By-Channel Convolutions *Oct. 2023*

Honors & Awards

Awards

Oct. 2024 **Excellent Prize (3rd Place)**, National Cryptography Contest, NSRI *(for [C01])*

Oct. 2023 **Special Prize**, National Cryptography Contest, NSRI *(for [J01])*

Dec. 2019 **First Place Scholarship**, Software-Focused University Program (Undergraduate)

Honors & Scholarships

2025–2026 **SNU Teaching and Research Support Scholarship**, Seoul National University

2022–2024 **BK 21+ Scholarship**, Ministry of Education of Korea

2018–2021 **National Scholarship for Outstanding Students in Science & Engineering**, Korea Student Aid Foundation

Experiences

Teaching

Seoul National University *Seoul, Republic of Korea*

Teaching Assistant, Department of Mathematical Sciences *Sep. 2021 - Present*

- Introduction to Cryptography (2025 Winter)
- Calculus for Humanities and Social Sciences 1 (2024 Winter)
- Calculus Practice 2 (2021, 2023 Winter)
- Calculus Practice 1 (2023, 2025, 2026 Spring)
- Differential Equation and Practice (2022 Winter)
- Linear Algebra (2022 Spring)
- Linear Algebra 2 (2026 Winter)

Internships

CryptoLab *South Korea*

Research Intern (Part-time) *Sep. 2025 - Feb. 2026*

CryptoLab and FHELab *France*

Research Intern (Full-time) *Jul. 2025 - Sep. 2025*

T-Brothers *Canada*

Undergraduate Intern *Jan. 2019 - Mar. 2019*

- Developed in-house applications for controlling the flow of stocks.

Education Program

University of Southern California (USC) *Los Angeles, USA*

SW Pioneer Program *Jul. 2019 - Aug. 2019*

Education

Integrated Ph.D. in Mathematical Sciences *Seoul, Republic of Korea*

Seoul National University *Sep. 2021 - Present*

- Advisor: Prof. Jung Hee Cheon (from Jan. 2023).
- Focus: Homomorphic Encryption, Privacy-Preserving Machine Learning.

B.S. in Mathematical Sciences and Industrial Mathmatics *Busan, Republic of Korea*

Pusan National University *Mar. 2015 - Aug. 2021*

- Summa Cum Laude